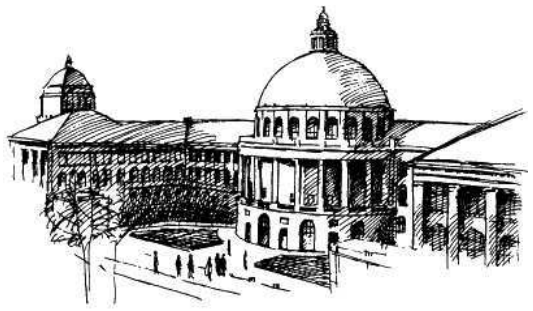


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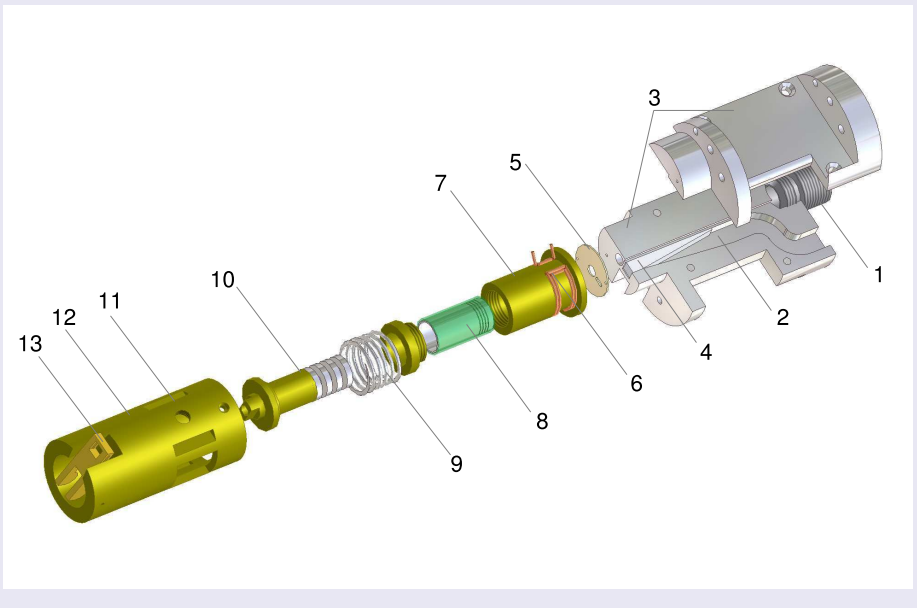
Universal Probe Head for EPR at 35 GHz

I. Gromov, J. Forrer and A. Schweiger

Design

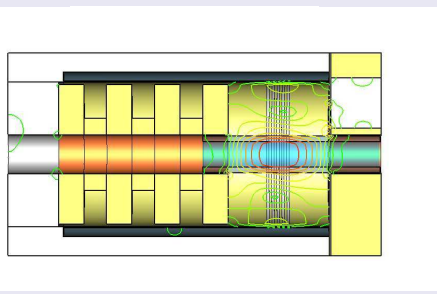
Since Q/Ka band pulse EPR spectrometers became routine spectroscopic tools in physics, chemistry and biology [1-3], the construction of probe heads for daily use on such instruments is a demanding task for the designer. Performance, sensitivity and convenience in use are three major criteria for which an optimal relation has to be reached. We present a probe head which is designed for a 100-W pulse Q-band spectrometer [3] equipped with a CF935 cryostat (Oxford). The probe head is optimized for advanced pulse EPR experiments, pulse ENDOR and cw measurements, with the priority in this order. It is based on a cylindrical cavity (8) coupled on the top to a waveguide via a diaphragm (5) and a Gordon coupling structure (2) with movable teflon wedge (4). The coupling structure has a specially shaped waveguide extruded in a solid block (3), which serves as the base to assemble the holders (7,12), rf coaxes, matching RF circuits and tuning rods. It also provides a guide (1) for the sample holder to allow for a precise positioning of the sample tube (od up to 3.9 mm) into the cavity. The TE_{011} cavity mode is used for the $B_1 \perp B_0$ configuration and the TE_{112} mode could be optionally utilized for $B_1 \parallel B_0$ measurements. However the cavity assembly is optimized to work in the TE_{011} mode. The cavity (8) is made of a quartz tube plated with Au/Ag on its inner surface. On the middle of the resonating volume the metallic layer is circularly scratched to make it transparent to the radio frequency field used for ENDOR experiments. A four-turn saddle coil (6) is wrapped around the cavity housing (7) to produce this field. A movable short (10) allows for a frequency tuning of about 2 GHz for the TE_{011} mode. A lever (13) fixed on a housing (12) and a spring (9) allow the smooth movement of the short (10). Exchangeable modulation coils (not shown) are fixed in slits (11).

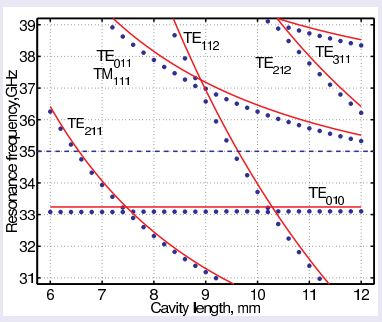
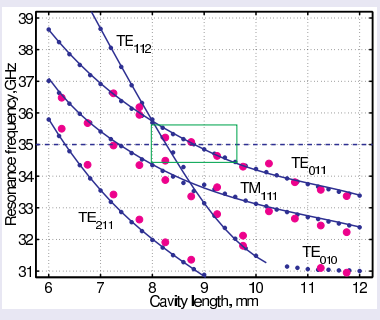
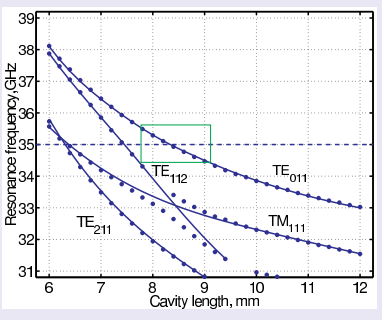
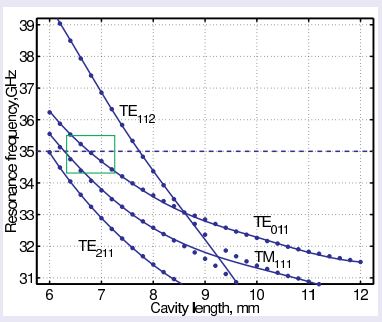
[1] I. Gromov, S. Orlinskii and R. Rakhmatullin, Appl. Magn. Reson, 3,1147 (1992)
[2] C. Davoust, P. Doan and B. Hoffman, J. Magn. Reson. A 119, 38 (1996)
[3] I. Gromov, J. Shane, J. Forrer, R.Rakmatullin, Yu. Rozentzwaig, A. Schweiger, J. Magn. Reson. 149, 196 (2001)



Simulations

The influence of the coupling structure, the entry holes , and the size and dielectric properties of the sample tube on the cavity eigenmodes and the emf distribution were simulated numerically using the MW Studio package (ver. 4.0, CST). The eigenfrequencies were calculated as a function of the cavity length to simulate the cavity behavior under operation, i.e. when one tries to get the proper frequency at different conditions. Based on these simulations the cavity radius of 5.5 mm has been chosen to avoid a TE_{011} /TE_{112} mode crossing in the working range of our spectrometer (34.5-35.5 GHz). The figures below show the eigenmodes of the cavity. Blue lines and dots represent numerical simulations, red lines are analytical solutions, magenta circles are experimental results, and the green squares are the operational areas in our case. As an example of an emf distribution, the isolines in the figure on the right show the amplitude of the magnetic fields of the TE_{011} mode inside the cavity with an 2.9/2 mm (od/id) empty quartz tube at 35 GHz.

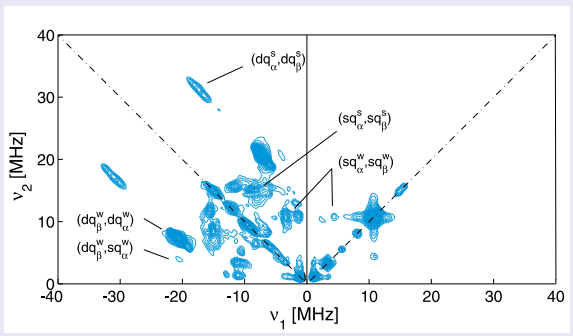


A. Empty cavity coupled to waveguide, no holes for samples	B. Cavity with 2.9/2 mm (od/id) empty quartz tube, ε_t=3.75	C. 2.9/2 mm quartz tube, ε_t=3.75, loading ε_s=2	D. 3.8/2.9 mm empty quartz tube, ε_t=3.75
			
The cavity shows several resonance modes. The numerical solutions (•) agree well with the analytical ones (–). The TE_{011} and TM_{111} modes are degenerated. The TE_{212}, TE_{311} and TE_{010} modes are shown for completeness. These modes do not cross the 35 GHz line under the studied conditions. The TE_{011}, TE_{211}, TM_{111} and TE_{112} modes have been observed during an exploitaion of the probe.	The TE_{011}/TM_{111} degeneracy is lifted due to the sample access holes and the sample tube. All modes are shifted to lower frequencies compared to case (A). The TE_{112} mode is shifted more than the TE_{011} mode. The magenta circles are experimental values. The TE_{112} mode crosses the line of the TE_{011} , but it shows only weak resonances because the coupling structure is optimized for the TE_{011} mode.	When the sample tube is filled with a dielectric material the TE_{112} mode is shifted further down in frequency compared to case (B) and with respect to the TE_{011} resonances. In the case at hand, the TE_{011} mode is always the first observed mode when the cavity length is reduced from 12 to 6 mm using a frequency in the range 33-38 GHz.	The wall of the sample tube in this case is closer to the electric field maxima of the TE_{011} mode than in cases (B,C). As a consequence the TE_{011} mode is shifted more to lower frequencies than the TE_{112} mode. Nevertheless, the TE_{011} mode is well separated from the TE_{112} mode and this situation is not changed when the tube is loaded with dielectric material.

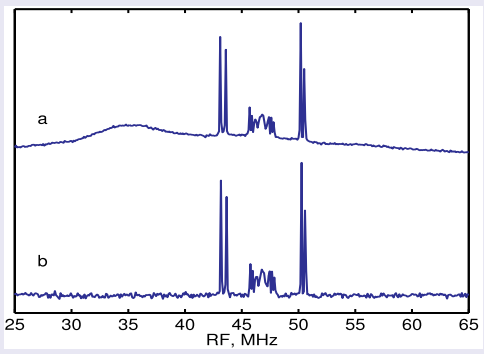
Performance

Parameter	Value	Conditions
Cavity radius, mm	5.5	
Active length, mm	6-10	
MW conversion factor, Gauss/$\sqrt{W}$	2.5	Q≈300
RF conversion factor, mT/$\sqrt{W}$	0.3	low Q tuned RF coil
Sensitivity, spins/$\sqrt{Hz}$	$3x10^9$	pulse mode, RT,Q≈150
Tuning range, GHz	34-36	
Temperature range, K	5-300	
Sample access, mm	3.9	
Q-factor	10^2-10^3	

Applications



Nitrogen HYSCORE spectrum of the MCRred2 form of methyl-coenzyme M reductase measured at the field position in the middle of the Ni(I) EPR spectrum at 15 K (C. Finazzo et.al., JBIC, 8, 586 (2003)). The single- and double-quantum cross-peaks from the weaker (w) and stronger (s) coupled nitrogens are observed. 16-ns mw pulses have been used to produce π/2 and π pulses in the HYSCORE sequence at 35.3 GHz.



Davies ENDOR spectra of protons weakly coupled to Cu(II) ions doped in a α-glycine single crystal. The spectra were recorded at a temperature of 15 K with an rf pulse length of 25 μs and an unmatched rf coil. The incident rf power was about 450 W. The spectrum (a) was measured in a standard way, i.e. the RF was swept linearly, at a repetition rate of 100 Hz . Because of the weakness of the continuous-flow cooling system the base line is heavily distorted. Applying a *random* RF frequency sweep using recently developed spectrometer software (SpecMan, ETH and WIS, see www.esr.ethz.ch) the baseline distortion is completely eliminated (b).